1. Launch one aws ec2 and install jenkins.
2. Setup Apache Tomcat on Ubuntu Virtual Machine
3. Setup Docker Desktop on your local windows.

Instances (1/1) Info

Connect | Instance state ▼ | Actions ▼ | Launch in

Find Instance by attribute or tag (case-sensitive) | All states ▼

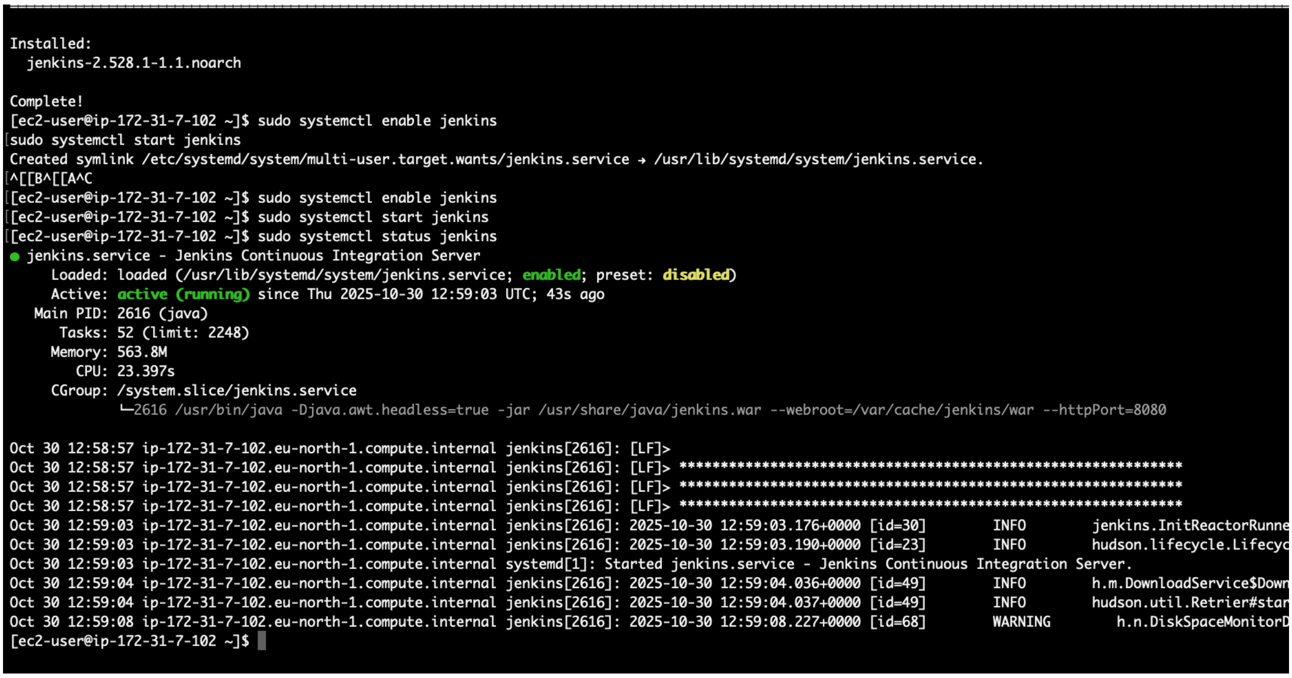
Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS
server tomcat	i-04691240af52436b7	Running	t3.small	3/3 checks passec	View alarms +	eu-north-1c	ec2-13-50-240-232.eu-...

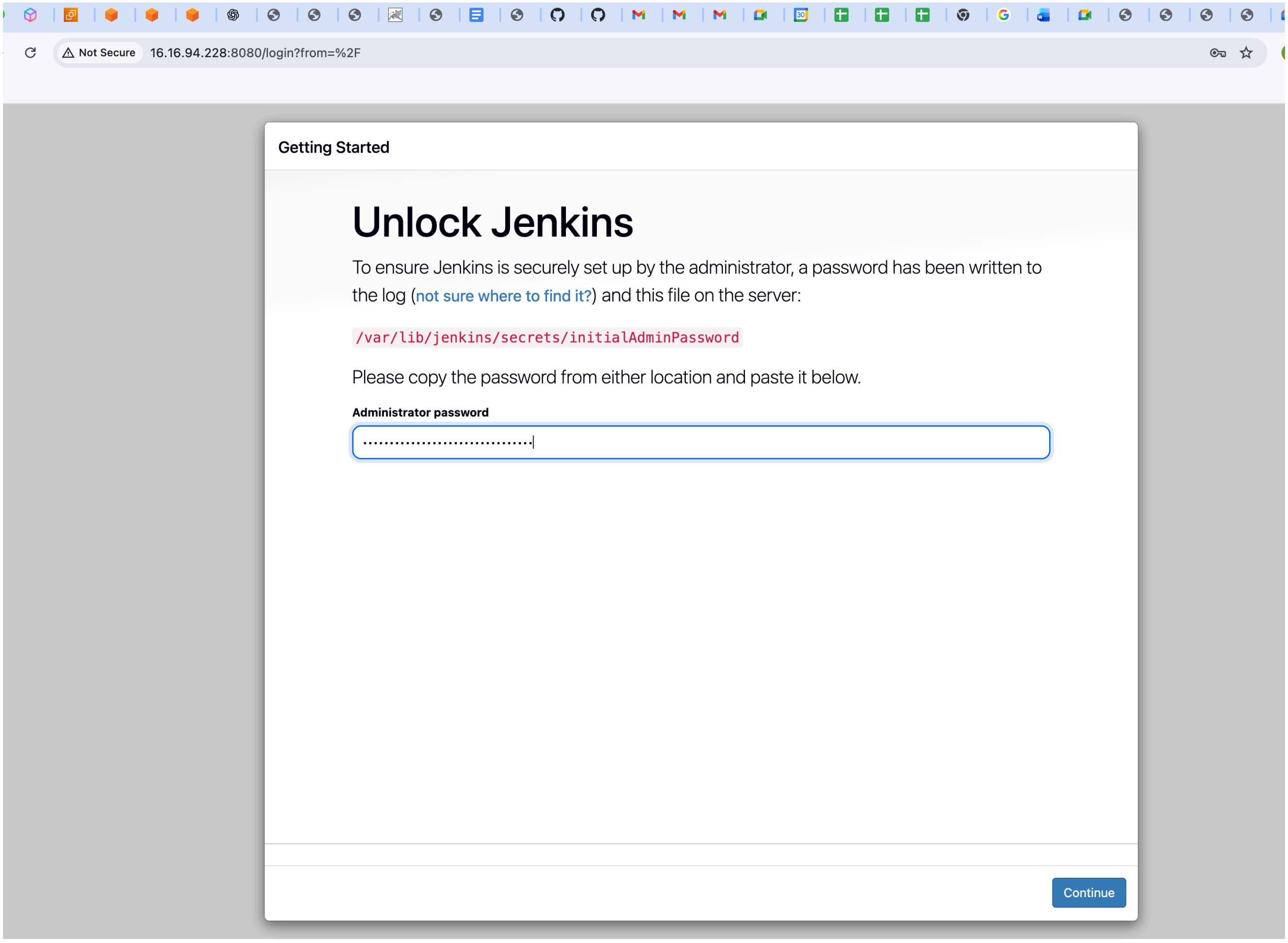
```
waseemakram@waseem downloads % ssh -i waseem.pem ec2-user@13.50.240.232


A newer release of "Amazon Linux" is available.
  Version 2023.9.20251020:
  Version 2023.9.20251027:
Run "/usr/bin/dnf check-release-update" for full release and version update info
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
Last login: Thu Oct 30 11:44:21 2025 from 13.48.4.203
[ec2-user@ip-172-31-7-102 ~]$ ls
```

Installed jenkins
By yum install Jenkins

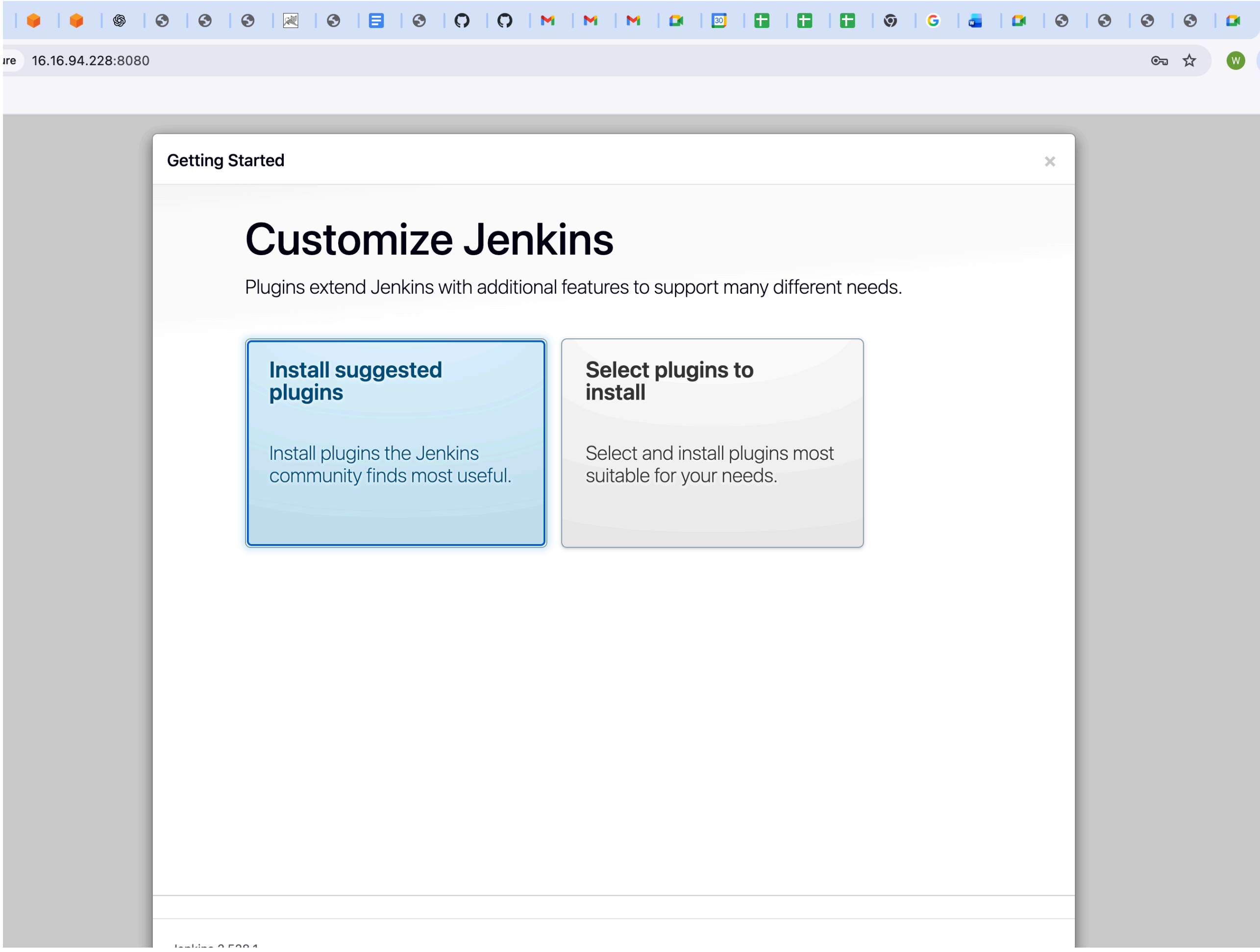
```
[ec2-user@ip-172-31-7-102 ~]$ sudo yum install jenkins -y
Jenkins-stable                                                                    615 kB/s |  32 kB
Dependencies resolved.
==========================================================================================================
 Package                     Architecture               Version                   Repository
==========================================================================================================
Installing:
 jenkins                     noarch                     2.528.1-1.1               jenkins

Transaction Summary
==========================================================================================================
Install  1 Package

Total download size: 91 M
Installed size: 91 M
Downloading Packages:
jenkins-2.528.1-1.1.noarch.rpm                                                     36 MB/s |  91 MB
----------------------------------------------------------------------------------------------------------
Total                                                                              35 MB/s |  91 MB
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing        :
  Running scriptlet: jenkins-2.528.1-1.1.noarch
  Installing       : jenkins-2.528.1-1.1.noarch
  Running scriptlet: jenkins-2.528.1-1.1.noarch
  Verifying        : jenkins-2.528.1-1.1.noarch
==========================================================================================================
WARNING:
  A newer release of "Amazon Linux" is available.

  Available Versions:

  Version 2023.9.20251020:
    Run the following command to upgrade to 2023.9.20251020:

      dnf upgrade --releasever=2023.9.20251020

    Release notes:
     https://docs.aws.amazon.com/linux/al2023/release-notes/relnotes-2023.9.20251020.html

  Version 2023.9.20251027:
    Run the following command to upgrade to 2023.9.20251027:
```

Sudo systemctl start Jenkins



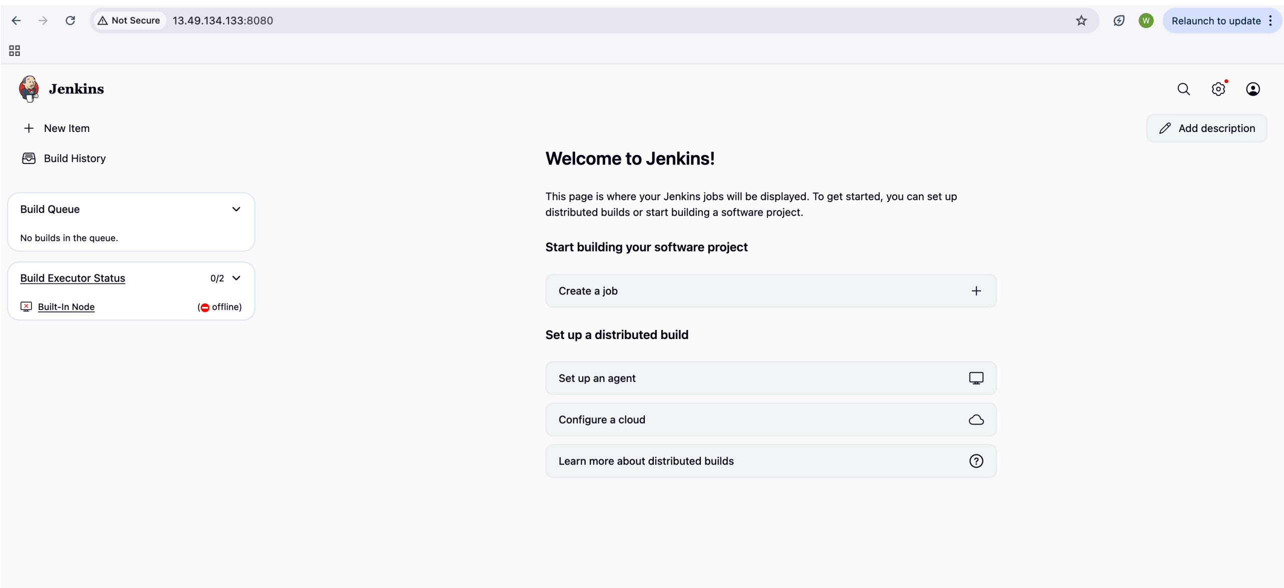


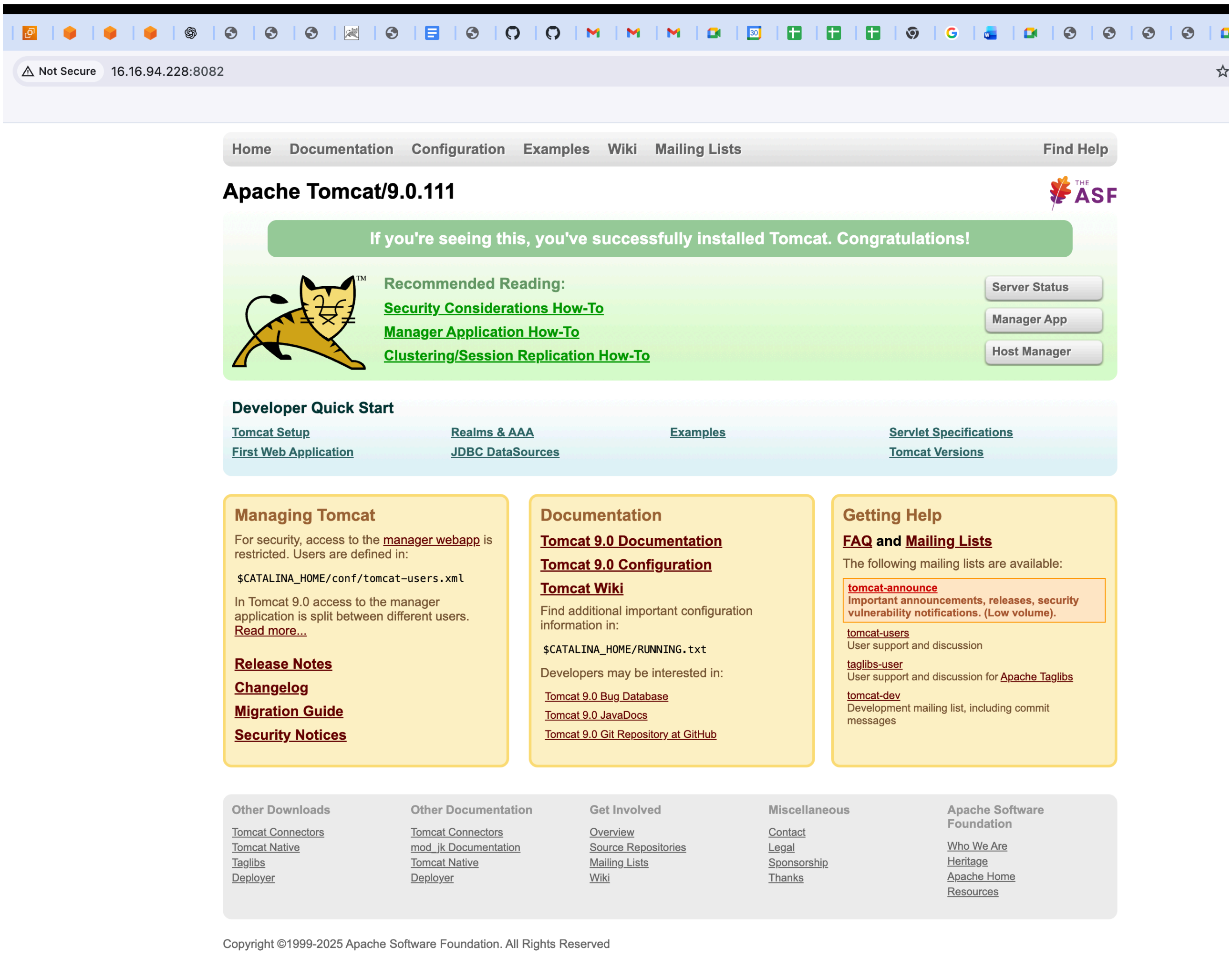
Cat password of Jenkins and copied and past to Jenkins page



Cat command for viewing password

```
cat: /var/lib/jenkins/secrets/initialadminpassword: No such file or directory
[ec2-user@ip-172-31-7-102 ~]$ sudo cat /var/lib/jenkins/secrets/initialAdminPassword
3c3a5c725dcd4430b80d87060e191af5
[ec2-user@ip-172-31-7-102 ~]$
```





Ps aux | grep tomcat for checking

```
[ec2-user@ip-172-31-7-102 ~]$ ps aux | grep tomcat
root        1465  0.4  7.1 3883928 140160 ?      Sl   12:43   0:06 /opt/java-17/bin/java -Djava.util.logging.config
file:/dev/./urandom -Djdk.tls.ephemeralDHKeySize=2048 -Djava.protocol.handler.pkgs=org.apache.catalina.webresources
th /opt/apache-tomcat-9.0.111/bin/bootstrap.jar:/opt/apache-tomcat-9.0.111/bin/tomcat-juli.jar -Dcatalina.base=/opt
ec2-user    3095  0.0  0.1 222328  2128 pts/0    S+   13:08   0:00 grep --color=auto tomcat
[ec2-user@ip-172-31-7-102 ~]$ sudo netstat -tulnp | grep 8080
tcp6       0      0 :::8080                 :::*                    LISTEN      2616/java
```

Yum install docker

```
-bash: docker: command not found
[ec2-user@ip-172-31-7-102 ~]$ sudo yum install docker -y
Last metadata expiration check: 0:25:35 ago on Thu Oct 30 12:58:07 2025.
Dependencies resolved.
================================================================================================================================
 Package                                   Architecture                        Version
================================================================================================================================
Installing:
 docker                                    x86_64                              25.0.13-1.amzn2023.0.1
Installing dependencies:
 container-selinux                         noarch                              4:2.242.0-1.amzn2023
 containerd                                x86_64                              2.1.4-1.amzn2023.0.1
 iptables-libs                             x86_64                              1.8.8-3.amzn2023.0.2
 iptables-nft                              x86_64                              1.8.8-3.amzn2023.0.2
 libcgroup                                 x86_64                              3.0-1.amzn2023.0.1
 libnetfilter_conntrack                    x86_64                              1.0.8-2.amzn2023.0.2
 libnfnetlink                              x86_64                              1.0.1-19.amzn2023.0.2
 libnftnl                                  x86_64                              1.2.2-2.amzn2023.0.2
 pigz                                      x86_64                              2.5-1.amzn2023.0.3
 runc                                      x86_64                              1.3.1-1.amzn2023.0.1

Transaction Summary
================================================================================================================================
Install  11 Packages

Total download size: 74 M
Installed size: 280 M
Downloading Packages:
(1/11): container-selinux-2.242.0-1.amzn2023.noarch.rpm
(2/11): iptables-libs-1.8.8-3.amzn2023.0.2.x86_64.rpm
(3/11): iptables-nft-1.8.8-3.amzn2023.0.2.x86_64.rpm
(4/11): libcgroup-3.0-1.amzn2023.0.1.x86_64.rpm
(5/11): libnetfilter_conntrack-1.0.8-2.amzn2023.0.2.x86_64.rpm
(6/11): libnfnetlink-1.0.1-19.amzn2023.0.2.x86_64.rpm
(7/11): libnftnl-1.2.2-2.amzn2023.0.2.x86_64.rpm
(8/11): pigz-2.5-1.amzn2023.0.3.x86_64.rpm
(9/11): containerd-2.1.4-1.amzn2023.0.1.x86_64.rpm
(10/11): runc-1.3.1-1.amzn2023.0.1.x86_64.rpm
(11/11): docker-25.0.13-1.amzn2023.0.1.x86_64.rpm
--------------------------------------------------------------------------------------------------------------------------------
Total
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing        :
  Installing       : runc-1.3.1-1.amzn2023.0.1.x86_64
  Installing       : containerd-2.1.4-1.amzn2023.0.1.x86_64
  Running scriptlet: containerd-2.1.4-1.amzn2023.0.1.x86_64
  Installing       : pigz-2.5-1.amzn2023.0.3.x86_64
  Installing       : libnftnl-1.2.2-2.amzn2023.0.2.x86_64
  Installing       : libnfnetlink-1.0.1-19.amzn2023.0.2.x86_64
  Installing       : libnetfilter_conntrack-1.0.8-2.amzn2023.0.2.x86_64
```

Docker started

```
Installed:
  container-selinux-4:2.242.0-1.amzn2023.noarch        containerd-2.1.4-1.amzn2023.0.1.x86_64        docker-2
  libnetfilter_conntrack-1.0.8-2.amzn2023.0.2.x86_64   libnfnetlink-1.0.1-19.amzn2023.0.2.x86_64     libnftnl

Complete!
[ec2-user@ip-172-31-7-102 ~]$ sudo systemctl start docker
[ec2-user@ip-172-31-7-102 ~]$ sudo systemctl enable docker
Created symlink /etc/systemd/system/multi-user.target.wants/docker.service → /usr/lib/systemd/system/docker.servi
[ec2-user@ip-172-31-7-102 ~]$ sudo systemctl status docker
● docker.service - Docker Application Container Engine
     Loaded: loaded (/usr/lib/systemd/system/docker.service; enabled; preset: disabled)
     Active: active (running) since Thu 2025-10-30 13:24:32 UTC; 1min 17s ago
TriggeredBy: ● docker.socket
       Docs: https://docs.docker.com
   Main PID: 5643 (dockerd)
      Tasks: 9
     Memory: 28.7M
        CPU: 348ms
     CGroup: /system.slice/docker.service
             └─5643 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock --default-ulimit nofil

Oct 30 13:24:32 ip-172-31-7-102.eu-north-1.compute.internal systemd[1]: Starting docker.service - Docker Applicat
Oct 30 13:24:32 ip-172-31-7-102.eu-north-1.compute.internal dockerd[5643]: time="2025-10-30T13:24:32.122147493Z"
Oct 30 13:24:32 ip-172-31-7-102.eu-north-1.compute.internal dockerd[5643]: time="2025-10-30T13:24:32.191855280Z"
Oct 30 13:24:32 ip-172-31-7-102.eu-north-1.compute.internal dockerd[5643]: time="2025-10-30T13:24:32.696565239Z"
Oct 30 13:24:32 ip-172-31-7-102.eu-north-1.compute.internal dockerd[5643]: time="2025-10-30T13:24:32.722868267Z"
Oct 30 13:24:32 ip-172-31-7-102.eu-north-1.compute.internal dockerd[5643]: time="2025-10-30T13:24:32.723032786Z"
Oct 30 13:24:32 ip-172-31-7-102.eu-north-1.compute.internal dockerd[5643]: time="2025-10-30T13:24:32.777975586Z"
```

Hello from docker

```
[ec2-user@ip-172-31-7-102 ~]$ sudo docker run hello-world
Unable to find image 'hello-world:latest' locally
latest: Pulling from library/hello-world
17eec7bbc9d7: Pull complete
Digest: sha256:56433a6be3fda188089fb548eae3d91df3ed0d6589f7c2656121b911198df065
Status: Downloaded newer image for hello-world:latest

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
 1. The Docker client contacted the Docker daemon.
 2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
    (amd64)
 3. The Docker daemon created a new container from that image which runs the
    executable that produces the output you are currently reading.
 4. The Docker daemon streamed that output to the Docker client, which sent it
    to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
 $ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
 https://hub.docker.com/

For more examples and ideas, visit:
 https://docs.docker.com/get-started/

[ec2-user@ip-172-31-7-102 ~]$
```